

Equations with Absolute Values

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Definitions

Absolute Values

Definitions

Properties of Absolute Value

Types of equations with absolute value

Type 1: $|A(x)| = c$

Type 2:

$$|A(x)| = c \cdot |B(x)|$$

Type 3:

$$|A(x)| + |B(x)| = c$$

Another Example

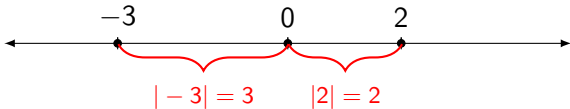
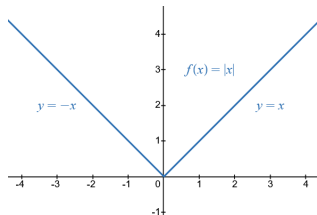
Definition:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

For example, $|5| = 5$, but
 $|-5| = -(-5) = 5$.

Intuition: Absolute value is the distance between x and 0 on the number line.

Graph of $f(x) = |x|$:



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Another Example

- ▶ $|x| \geq 0$ (Absolute value is always non-negative)
- ▶ $|x| = 0 \iff x = 0$
- ▶ $|ab| = |a||b|$ (Multiplicative property)
- ▶ $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$ ($b \neq 0$)
- ▶ $|x - y| = |y - x|$ (Symmetry)
- ▶ $||x|| = |x|$ (Idempotent)
- ▶ $\sqrt{x^2} = |x|$ (Square root of square)

Types of equations with absolute value

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Another Example

► **Type 1:** $|A(x)| = c$

Solve: $A(x) = c$ or $A(x) = -c$ Only valid if $c \geq 0$

► **Type 2:** $|A(x)| = c \cdot |B(x)|$

Rewrite as: $\left| \frac{A(x)}{B(x)} \right| = c$ Then solve: $\frac{A(x)}{B(x)} = \pm c$

► **Type 3:** $|A(x)| + |B(x)| = c$

Find critical points where inside of $A(x), B(x) = 0$

Split into intervals and remove absolute values based on sign

Always check your answer by substituting it back to the original equation

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Example. Solve for x : $|2x - 5| = 7$.

$$2x - 5 = 7 \quad \text{or} \quad 2x - 5 = -7$$

$$x = 6 \quad \text{or} \quad x = -1$$

The two solutions are then $x = 6, -1$.

Do It Yourself

$$|2x + 1| = 7 \quad \Rightarrow \quad x = 3, -4$$

Type 2: $|A(x)| = c \cdot |B(x)|$

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Example. Solve $|x - 2| = 3|x + 1|$.

$$\Rightarrow \left| \frac{x-2}{x+1} \right| = 3 \Rightarrow \frac{x-2}{x+1} = \pm 3$$

Case 1: $\frac{x-2}{x+1} = 3 \Rightarrow x = -\frac{5}{2}$

Case 2: $\frac{x-2}{x+1} = -3 \Rightarrow x = -\frac{1}{4}$

Answer:

$$x = -\frac{5}{2}, -\frac{1}{4}$$

Do It Yourself

$$|x + 1| = 4|3x - 8| \Rightarrow x = 3, \frac{31}{13}$$

Type 3: $|A(x)| + |B(x)| = c$

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[Another Example](#)

General Method:

- ▶ Find critical points where each expression inside absolute value is 0 i.e. $A(x) = 0, B(x) = 0$
- ▶ Split into intervals
- ▶ Remove absolute values based on sign in each interval
- ▶ Solve and validate

Type 3 continues

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Example. Solve $|x - 3| + |x + 1| = 8$.

Critical points: $x = -1, 3$

Case 1: $x < -1$

$$-(x - 3) - (x + 1) = 8 \Rightarrow -2x + 2 = 8 \Rightarrow x = -3 \quad (\text{Valid})$$

Case 2: $-1 \leq x < 3$

$$-(x - 3) + (x + 1) = 8 \Rightarrow 4 \neq 8 \quad (\text{No solution})$$

Case 3: $x \geq 3$

$$(x - 3) + (x + 1) = 8 \Rightarrow 2x - 2 = 8 \Rightarrow x = 5 \quad (\text{Valid})$$

Answer: $x = -3, 5$

Do It Yourself

$$|x - 4| + |x + 2| = 10 \Rightarrow x = -4, 6$$

Equation with $\sqrt{x^2}$

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Another Example

Recall that $\sqrt{x^2} = |x|$. For example, $\sqrt{(-6)^2} = \sqrt{36} = 6$.

Example.

Solve for u : $\frac{18}{u} = \sqrt{x^2} - 2$.

$$\frac{u}{18} = \frac{1}{|x| - 2} \implies \boxed{u = \frac{18}{|x| - 2}}$$